DHBW Ravensburg

# Intelligence Engineering

## Solving Real World Problems At Scale

Prof. Dr.-Ing. Mark Schutera | November 5, 2026 · *drifting stream*

LECTURE _

# Orientation

what this course is, what it assumes, what you are going to know afterwards.

You have been given a lot of raw **data**,

# and you want to engineer the shit out of it.

You have been given a business problem,

# and you want to choose the right metrics and objectives to solve it.

Your initial models perform well in offline experiments,

# and you want to deploy them at scale.

You want to harden your deployments

# to quickly detect, debug, and address issues that arise in production.

There is a lot of manual work and handling involved so far,

# and you want an operation vacation for your machine learning product or service.

You want to learn about a framework and foundation for building machine learning systems

# that can be shared and reused across machine learning use cases and data products.

1

# Overview

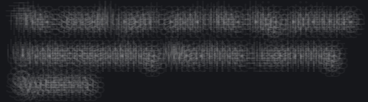
## Machine Learning and Data Engineering

# The lecture in one slide

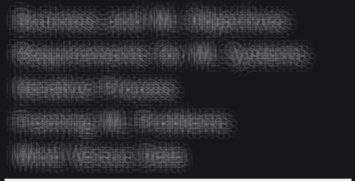
| Lecture | Contents |
|---|---|
| I | Overview of Machine Learning Systems |
| II | Introduction to Machine Learning Systems Design |
| III | Data Engineering Fundamentals |
| IV | Training Data |
| V | Feature Engineering |
| VI | Model Development and Offline Evaluation |
| VII | Model Deployment and Prediction Service |
| VIII | Data Distribution Shifts and Monitoring |
| IX | Continual Learning and Test in Production |
| X | Infrastructure and Tooling for MLOps |
| XI | The Human Side of Machine Learning |

# The lecture in one slide

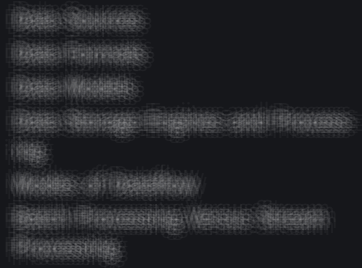
**I Overview of Machine Learning Systems**



**II Introduction to Machine Learning Systems Design**

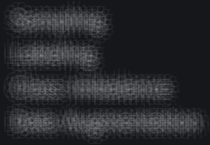


**III Data Engineering Fundamentals**



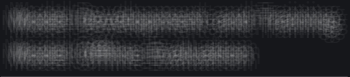
# The lecture in one slide

## IV Training Data



## V Feature Engineering

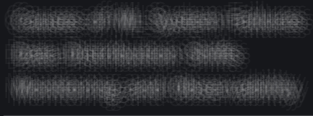
## VI Model Development and Offline Evaluation



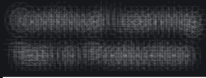
# The lecture in one slide

**VII Model Deployment and Prediction Service**

**VIII Data Distribution Shifts and Monitoring**

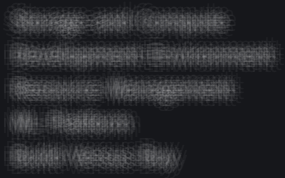


**IX Continual Learning and Test in Production**

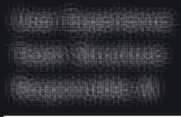


# The lecture in one slide

**X Infrastructure and Tooling for MLOps**



**XI The Human Side of Machine Learning**



# Course Work

The Hive is Real. Dataset on Hugging Face.

**Presentation (15 min) + Q&A (open)**

You get recordings of a beehive entrance and contribute data engineering, an annotation policy, a prediction service or specialized tooling.

Open whiteboard, three slides or one page only. Be ready when the class starts.

The bee

# Course Work

What I am looking for:

**I**ntroduction by Motivation and Literature

**M**ethod by Explanation and/or Demonstration

**R**esults show what was achieved and how well

**D**iscussion on findings, limitations and what's next

**Smart and Aesthetic** Thoughts, Designs, Ideas and Questions.

The bee

# Extra Mile

**Special projects**

If you are up for the extra mile, let me know. I have special projects that I can assign to you.

Wrapped in the same presentation and Q&A format. Yes, it will be more challenging and take more time and effort.

A classic: lose-lose in the short term, yet **win-win in the long term**.

A large banana

# Brainwriting, 90 minutes

1. **5 min** Form groups of **4 to 5** students.
2. **10 min** Research: Google, Agent, you name it.
3. **10 min** Each student writes four ideas.
4. **5 min** Research.
5. **10 min** Stacks pass to the next table, each adds two.
6. **5 min** Research.
7. **10 min** Stacks pass again, each adds the last two.
8. **15 min** Cluster and clean the ideas per group.
9. **15 min** Cluster across teams.
10. **5 min** Walk the idea deck: everyone looks at all clusters.

The bee

It is your obligation to

# get yourself a presentation slot and brainstorm a project idea to work on.

LECTURE I

# Overview of a Machine Learning System

# This lecture in one slide

**I Overview of Machine Learning Systems**

The small part and the big picture

Understanding Machine Learning Systems

A **model** is only a small part from the system around it, and some might argue there are a lot of moments when not to reach for machine learning at all.

1

# The small part and the big picture

## Don’t confuse the model with the machine learning system.

# Enabling the delivery of the logic

The **model** is the small part of a machine learning system. With foundation models and AI engineering, that part got smaller still. The essential work of a machine learning system remains and stacks around business understanding, infrastructure, design pattern and data engineering.

Before learning about solving a problem with machine learning, let's first assess when it is actually a good idea to do so.

# (1) Learn

The problem you are solving, requires your system to have the capability to **learn**. Learning can look different depending on the problem and its context and boundary conditions. You already know about supervised learning, unsupervised learning and reinforcement learning paradigms.

Machine learning is an approach to (1) learn (2) complex patterns from (3) existing data and use these patterns to make (4) predictions on (5) unseen data.

# (2) Pattern

Learning requires the existence of a **pattern**, as opposed to noise or randomness. It is not trivial nor obvious whether there is a pattern in the data, and whether it is learnable. Complex pattern is more difficult to embed or state explicitly, this is where machine learning has an edge over classical approaches.

Machine learning is an approach to (1) learn (2) complex patterns from (3) existing data and use these patterns to make (4) predictions on (5) unseen data.

# (3) Data

Data is available, accessible or acquirable. The data is of **sufficient** quality and quantity to support learning, (and to validate for that matter). If your approach does not require data, it is because you build on embeddings or foundation models, which themselves are built on data in the first place, or you need to rely on data acquisition and curation as part of your system.

Machine learning is an approach to (1) learn (2) complex patterns from (3) existing data and use these patterns to make (4) predictions on (5) unseen data.

# (4) Predictions

Machine learning is about making predictions, **always**. Predictions can be of different types, such as classification, regression, ranking, clustering, generation and on and on. By reframing problems in terms of predictions, we can leverage machine learning to solve them. Predictions are approximations by nature and therefore come with uncertainty, which needs to be accounted for.

Machine learning is an approach to (1) learn (2) complex patterns from (3) existing data and use these patterns to make (4) predictions on (5) unseen data.

# (5) Unseen Data

In a closed system or domain, the data is finite and the system can be fully specified by optimization. However, often the domain is open and infinite, and the system needs to be able to **generalize** to unseen data. Of course, the unseen data is not completely unknown, it is drawn from the same distribution (i.i.d.) as the training data and is thus able to generalize to the unseen data inside a specific domain or distribution.

Machine learning is an approach to (1) learn (2) complex patterns from (3) existing data and use these patterns to make (4) predictions on (5) unseen data.

# Catalysts

A catalyst is a substance that increases the rate of reaction without undergoing change itself.

**The task is repetitive.**

In a sense that we have a lot of data to learn a pattern from.

**It scales.**

The need to scale a specific prediction capability, not only a few times.

**The context changes.**

Meaning that hard coded or other solutions will be brittle and break.

**Being wrong is not too bad.**

Approximations on unseen data. Failures need to be manageable.

# When Not to Use Machine Learning

**It is unethical.**

Well, that might also be a general rule. But doing stuff at scale, without human oversight, has its implications.*

**There's a simple solution.**

A simple solution, heuristic, or baseline should always be the first step regardless.

**It's not economically viable.**

The value of the prediction, needs to be greater than the cost of the system, including development, deployment and maintenance.

* More? Enroll to my Philosophy of AI course.

Describe two use-cases and the rationale for whether or not to use machine learning.

# One Machine Learning Use-Case and One which is not.

# The field is moving, 2026

Adoption is near universal, value capture is not.

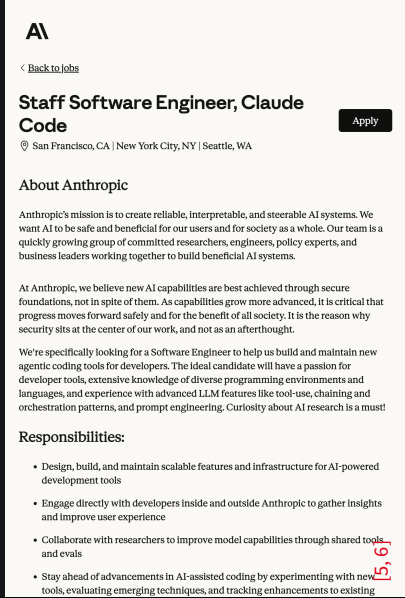
- **Adoption** 88% of organizations use AI in at least one function, 72% use gen AI.
- **Reach** Gen AI reached 53% of the population in three years, faster than the PC or the internet.
- **Productivity** 80% of users report personal gains.
- **Slop** 40% of desk workers say they received AI workslop in the past month. Each case costs almost two hours to fix, about $186 per employee per month.
- **Value** Only **37%** of organizations attribute any EBIT impact to AI, flat since 2025, and 6% qualify as high performers.

[2–4]

“… in twelve months, we may be in a world where AI is writing **essentially all of the code**.”

**Dario Amodei, March 2025. Nineteen months later, Anthropic lists 94 open software engineer roles.**

Anthropic hiring, October 2026



Back to jobs

**Staff Software Engineer, Claude Code**

Apply

San Francisco, CA | New York City, NY | Seattle, WA

About Anthropic

Anthropic's mission is to create reliable, interpretable, and steerable AI systems. We want AI to be safe and beneficial for our users and for society as a whole. Our team is a quickly growing group of committed researchers, engineers, policy experts, and business leaders working together to build beneficial AI systems.

At Anthropic, we believe new AI capabilities are best achieved through secure foundations, not in spite of them. As capabilities grow more advanced, it is critical that progress moves forward safely and for the benefit of all society. It is the reason why security sits at the center of our work, and not as an afterthought.

We're specifically looking for a Software Engineer to help us build and maintain new agentic coding tools for developers. The ideal candidate will have a passion for developer tools, extensive knowledge of diverse programming environments and languages, and experience with advanced LLM features like tool-use, chaining and orchestration patterns, and prompt engineering. Curiosity about AI research is a must!

Responsibilities:

- Design, build, and maintain scalable features and infrastructure for AI-powered development tools
- Engage directly with developers inside and outside Anthropic to gather insights and improve user experience
- Collaborate with researchers to improve model capabilities through shared tools and evals
- Stay ahead of advancements in AI-assisted coding by experimenting with new tools, evaluating emerging techniques, and tracking enhancements to existing

[5, 6]

2

# Understanding Machine Learning Systems

## Research and Production are not the same.

# Horizontal versus Vertical

Nothing motivates system thinking more than the need to go horizontal. Academia is vertical, with a narrow focus on a single problem, where drilling deep, fast is rewarded and focus allows to teach a wide range of concepts.

| | Research | Production |
|---|---|---|
| **Objectives** | State of the art, highly engineered | Stakeholder outcome driven |
| **Compute** | Training and high throughput | Fast inference with low latency |
| **Data** | Often static benchmark oriented | Shifting domains and structures |
| **Usability metrics** | Usually not considered | UI, Fairness, Robust, Interpretability |

# Stakeholders and Objectives

In research the objective is one number on one benchmark. Ship the same model as a product and five people want five different things from it: **conflicting requirements** are the norm, and not all of them are must haves.

ML engineers

Platform team

Product team

Sales team

Manager

The bee hive

Different stakeholders have different objectives and think in different solution spaces.

# Define the next step for the bee hive and line out how to reach that.

# Five people, five solution spaces

Each of them is right. The **objective** is what they are measured on, the solution space is where they will look for it.

| | Objective | Solution space |
|---|---|---|
| ML engineers | better bee counts | more labelled frames, a bigger model |
| Platform team | nothing pages at 3am | freeze the model, build the platform |
| Product team | the hive view feels live | a latency budget of 100 ms |
| Sales team | revenue per hive | an alert for every hive, sold as premium |
| Manager | margin | fewer models, and maybe fewer engineers |

# Computational priorities

Research optimises **throughput**, production optimises **latency**.

| Processing | Latency | Throughput |
|---|---|---|
| One query at a time | 10 ms | 100 queries/s |
| Batches of 10 | 20 ms | 500 queries/s |
| Batches of 100 | 50 ms | 2,000 queries/s |

Every row is the same model on the same hardware. Only the batch changed.

Batching waits for the queue to fill, so each single request waits longer while the machine as a whole serves far more of them.

Batch size is not the only lever. Model size, sample size and how many models sit in one request all move the numbers.

# Latency is a distribution

Ten requests, latency in ms:

10.0, 9.7, 10.4, **90.0**, 9.9, 10.0, 11.0, 9.5, 10.2, 10.0

The mean would be 18.1 ms. Nine of the ten requests were served in about 10 ms. Mean only holds at larger scales.

2017, Akamai: A 100 ms delay can cost 7% of conversions.

2019, Booking.com: about 30% more latency cost about 0.5% in conversion.

2016, Google: Over half of mobile users leave a page that takes more than three seconds.

[6-7]

# Percentiles, not averages

The same ten requests, sorted. A percentile is a kind of promise: $q$ percent of requests finish in $p_q$ or less.

9.5, 9.7, 9.9, 10.0, 10.0 ($p_{50}$), 10.0, 10.2, 10.4, 11.0 ($p_{90}$), **90.0** ($p_{99}$)

$$p_q = \min\left\{ t : \frac{\#\,\text{requests} \le t}{n} \ge \frac{q}{100} \right\}$$

The mean of 18.1 ms moves with the outlier, the percentile makes the outlier visible.

With ten requests $p_{99}$ is just the maximum. It takes 100 requests before $p_{99}$ and the maximum can differ. Precision needs to be bought with data.

Quality gates count in nines: Three nines is $p_{99.9}$ (1 in 1000 requests), five nines $p_{99.999}$ (1 in 100000).

[10]

# Your toxic relationship with mean accuracy

Latency is one metric. It might not be yours to measure. Latency is a distribution. As is any metric. So if you are only reporting mean accuracy, you are **underdelivering** already.

Start with the feature the solution has to deliver on. Pick the metrics that speak to them. Then curate the data that allows to measure them. Only then build the methods that move those metrics.

[11]

# Data in research

A research dataset is a teaching artefact. Somebody already did the work, for something very specific.

- Clean and well formatted
- Static, so the community can benchmark against it
- Quirks of the dataset are known and documented
- Open source scripts to feed your models
- Historical, already stored somewhere
- Reported results usually overfit

Titanic [12]: 891 rows, 12 columns.

Karpathy: At Tesla you lose sleep over data

[13]

# Leaderboards and Benchmarks

Well curated data is expensive and cumbersome. Benchmarks allow comparability, come at scale, and usually link proper metrics and state of the art methods. And they can be adjusted.

- System This scores a model, and a model is not a system.
- Metric It measures not necessarily what you care about.
- Overfit Every run against a benchmark stales it, fast.
- Chance Seeds move CIFAR 10 by 1.82 points.
- Problem Benchmarks are a great proxy, but usually not your problem exactly.

GLUE 2018. Trust me bro, nine tasks

**GLUE: A Multi-Task Benchmark and Analysis Platform for Natural Language Understanding**

**Alex Wang**[1], **Amanpreet Singh**[1], **Julian Michael**[2], **Felix Hill**[3], **Omer Levy**[2], **and Samuel R. Bowman**[1]

[1]New York University, New York, NY
[2]Paul G. Allen School of Computer Science & Engineering, University of Washington, Seattle, WA
[3]DeepMind, London, UK

{alexwang,amanpreet,bowman}@nyu.edu
{julianjm,omerlevy}@cs.washington.edu
felixhill@google.com

Human ability to understand language is *general, flexible*, and *robust*. In contrast, most NLU models above the word level are designed for a specific task and struggle with out-of-domain data. If we aspire to develop models with understanding beyond the detection of superficial correspondences between inputs and outputs, then it is critical to develop a unified model that can execute a range of linguistic tasks across different domains.

To facilitate research in this direction, we present the General Language Understanding Evaluation (GLUE, gluebenchmark.com): a benchmark of nine diverse NLU tasks, an auxiliary dataset for probing models for understanding of specific linguistic phenomena, and an online platform for evaluating and comparing models. For some benchmark tasks, training data is plentiful, but for others it is limited or does not match the genre of the test set. GLUE thus favors models that can represent linguistic knowledge in a way that facilitates sample-efficient learning and effective knowledge-transfer across tasks. While none of the datasets in GLUE were created from scratch for the benchmark, four of them feature privately-held test data, which is used to ensure that the benchmark is used fairly.

We evaluate baselines that use ELMo (Peters et al., 2018), a powerful transfer learning technique, as well as state-of-the-art sentence representation models. The best models still achieve fairly low absolute scores. Analysis with our diagnostic dataset yields similarly weak performance over all phenomena tested, with some exceptions.

**The GLUE benchmark** GLUE consists of nine English sentence understanding tasks covering a broad range of domains, data quantities, and difficulties. As the goal of GLUE is to spur development of models that can quickly adapt to and multi-task learning (Caruana, 1997) and pretraining (Dai and Le, 2015; Peters et al., 2018; Radford et al., 2018; Howard and Ruder, 2018) are two promising techniques that leverage transfer learning to large amounts of unlabeled data. For the last of these techniques, we expect that the choice of pretraining objective will be especially important, with variants of the language modeling objective (Radford et al., 2018) likely to be well-suited to learning representations that capture properties of negation, but other objectives such as discourse coherence (Nie et al., 2018) may also show promise. We also expect that the relative performance of pretraining models will be highly dependent on additional training data and task formulation. We hope that GLUE will provide a productive framework for evaluating and analyzing models along these axes.

| Corpus | \|Train\| | Task | Domain |
|---|---|---|---|
| Single-Sentence Tasks | | | |
| **CoLA** | 8.5k | acceptability | misc. |
| SST-2 | 67k | sentiment | movie reviews |
| Similarity and Paraphrase Tasks | | | |
| MRPC | 3.7k | paraphrase | news |
| STS-B | 7k | textual sim. | misc. |
| **QQP** | 364k | paraphrase | online QA |
| Inference Tasks | | | |
| **MNLI** | 393k | NLI | misc. |
| QNLI | 108k | QA/NLI | Wikipedia |
| RTE | 2.5k | NLI | misc. |
| **WNLI** | 634 | coref./NLI | fiction books |

Table 1: Task descriptions and statistics. **Bold** denotes tasks for which there is privately-held test data. All tasks are binary classification, except STS-B (regression) and MNLI (three classes).

quire models to share substantial knowledge (e.g., trained parameters) across tasks, while maintaining some task-specific components. Though it is possible to train a model per task and evaluate the resulting set of models on this benchmark, we expect that inclusion of several data-scarce tasks will ultimately render this approach uncompetitive.

The nine tasks include two tasks with single-sentence inputs: Corpus of Linguistic Acceptability (CoLA; Warstadt et al. 2018) and Stanford Sentiment Treebank (SST-2; Socher et al. 2013). Three tasks involve detecting semantic similarity: Microsoft Research Paraphrase Corpus (MRPC; (Dolan and Brockett, 2005)), Quora Question Pairs[1] (QQP), and Semantic Textual Similarity Benchmark (STS-B; Cer et al. 2017). The remaining four tasks are formatted as natural language inference (NLI) tasks, such as the Multi-Genre NLI corpus (MNLI; Williams et al. 2018) and Recog-

# Data in production

Nobody did the work. The data is a by-product of a business that was not run for your benefit. Usually you don't get away with building and curating your own dataset to gain back control and clear the fog.

- Messy, noisy, possibly unstructured
- Likely biased, and you need to find out how
- Labels sparse, imbalanced, or incorrect
- Changing requirements can mean relabeling
- Privacy and regulatory concerns
- Generated by users, systems, third parties

# LLM as a judge (is the new ”I tested on the training data”)

An LLM judge grades your model with another model. Against its labels $\tilde{y}$ instead of the true $y$, your accuracy is off by as much as the judge is wrong.

Never for **validation**. For training data at most, and only while it adds no more $\tilde{y}$ than your ground truth already has.

- **Hallucination** grades claims it cannot check
- **Position bias** swap the answers, 65% of verdicts hold
- **Self-preference** rates its own, and its family’s, outputs higher

The judge meets the judged [17–19]

# Perpetuating Bias

Machine learning algorithms do not predict the future, they encode the past, and then they act on it.

**The invisible hand** of performative prediction plays out t scale: With an increase in agentic systems the prediction is not merely a reading of the world, it is a push on it.

2014 to 2018, Amazon built a résumé ranker on ten years of its own hiring. It learned to penalise the word “women’s” and graduates of two women’s colleges. The project was scrapped.

2015, Google Photos labelled two Black users as gorillas. The fix shipped, and three years later, was still to delete the label from the vocabulary.

[20–23]

# Interpretability

Research rewards performance, not interpretability. Industry asks for it, but is anyway slow to build it:

- **Users**, business leaders and end users alike, to see why a decision was made, so they can trust it and catch what is wrong with it.
- **Developers**, to debug, improve and align the model.

Most of us use a fridge without understanding it. Fewer feel that way about the thing deciding on their mortgage. Only 19% of large companies are working on explainability. Jev, 2026, is built only to decide: a typed answer with a probability, and no text to read an explanation from.

Press 2: Hand Claude recursive permission

[24]

# Human on the loop

Putting a human on the loop is the standard answer to an uninterpretable or weak model. The evidence that it works is **thin**.

Supervising reliable automation degrades exactly the capability being supervised: Vigilance falls, response to the first failure is poor, and the takeover you designed for arrives too late.

Tempe, 2018: the system saw the pedestrian 5.6 s before impact and handed over at 1.2 s. The safety driver, eyes on her phone for 34% of the drive, looked up 1 s before. Test a human on the loop like any other design decision.

Tempe Police dashcam, March 2018

# Why this is different from traditional software engineering

Traditional software engineering tests and versions code. A machine learning system has to test and version code, data, and everything the two produce together.

- How do you version a dataset, and diff two versions of one?
- How do you deploy into an environment that keeps moving?
- How do you monitor and debug a system with no stack trace?
- How do you avoid being harmful at scale and even push it?

Revisit the craft in **Full Stack Handwerkszeug**: Git, development and production, CI/CD.

Two books worth owning,

**Software Engineering at Google** [26]: engineering is programming integrated over time. Every question on the left is a question about time.

**The Mythical Man-Month** [27]: essential complexity against accidental. Data is essential complexity, and there is no silver bullet for it either.

**A trained model is the small box in the middle.**

You might argue that the engineering and development side of machine learning is more exciting.

**Most jobs will be, and already are in productionizing machine learning systems or decision making.**

LECTURE II

# Introduction to Machine Learning Systems Design

# This lecture in one slide

**II Introduction to Machine Learning Systems Design**

Business and ML Objectives

Requirements for ML Systems

Iterative Process

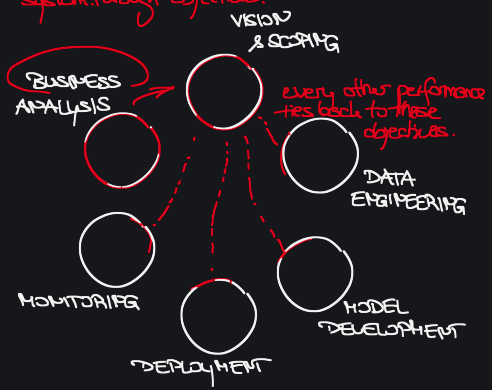
Framing ML Problems

Mind Versus Data

1

# Business and ML Objectives

First Guidance on Developing the System through Objectives.




"The sole purpose of a business is to maximize profits for shareholders" - Milton Friedman

VISION & SCOPING
BUSINESS ANALYSIS
DATA ENGINEERING
MODEL DEVELOPMENT
DEPLOYMENT
MONITORING

There is vested interest in the AI hype. Expectation management is crucial.
VISION & SCOPING
BUSINESS ANALYSIS
Starting next AI adoption is hard work and an investment.
DATA ENGINEERING
MONITORING
MODEL DEVELOPMENT
DEPLOYMENT
Returns on investment heavily depend on maturity of adoption.
Development cycles and skills don't develop over-time.

2

# Requirements for ML Systems

- **RELIABILITY**
  Correct function at desired level of performance in the face of adversity.
- **SCALABILITY**
  With respect to complexity, throughput, and other types of growth.
- **MAINTAINABILITY**
  Structure workload, infrastructure and tool landscape so that you can keep up.
- **ADAPTABILITY**
  Capacity to discuss potentials for performance improvement and allowing updates without service interruptions.

3

# Iterative Process

Developing ML systems is a cycle with a lot of back and forth!
#ITERATIVE.
VISION & SCOPING
BUSINESS ANALYSIS
DATA ENGINEERING
MODEL DEVELOPMENT
DEPLOYMENT
MONITORING

Objectives
Key Results
Constraints
VISION & SCOPING

Stakeholders: users, managers, humanity

We learn from data, so this is where we start with our craftmanship.
Different services, modalities, formats. - for examples
DATA ENGINEERING

It is all about the labels

Training Data
X
This requires the most ML specific knowledge.
ML
MODEL DEVELOPMENT

Training Data
X
This requires the most ML specific knowledge.
ML
MODEL
DEVELOPMENT
• Feature Engineering
• Model Selection
• Training
• Evaluation
ŷ

You will never reach a point where your system is done.
At some point you need to expose your system.
DEPLOYMENT

Monitoring: stay adaptive

Business analysis closes the loop

4

# Framing ML Problems

A lot can be framed as an ML problem

Classification versus regression

CLASSIFICATION
REGRESSION
0.2
0.8
0.3
class 1
class 2
Goal to enforced classes in X.
Threshold to class
Usually you want to reframe in this direction.
binary
multiclass
#Cardinality
Reframe in this direction if possible.

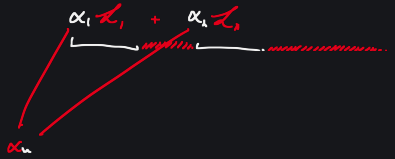
!! Restrain from „buying" this with an increasing number of models.
Hierarchical & Sequential systems are sophisticated but hard to maintain.

Talk is cheap. Go and try those things on MNIST to sharpen your own intuition.

Talk is cheap, sharpen your intuition on MNIST

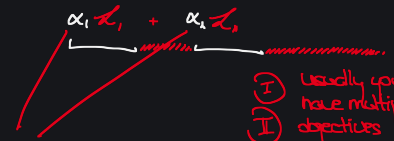
The objective function defines what optimal is

THE EFFECTIVENESS OF DECOUPLING OBJECTIVES.



THE EFFECTIVENESS OF DECOUPLING OBJECTIVES.

THE EFFECTIVENESS OF DECOUPLING OBJECTIVES.




I) Usually you have multiple objectives anyway.

II) This is a clean way to state them explicitly.

# Pareto Optimization

Trivial to weight objectives while training.

Everytime you touch is a retrain.

Beautiful opportunity to train two models and adaptively weight during inference.

! Objective Functions require gradients.
For everything else we still need metrics.

5

# Mind Versus Data

As data flows through models. During the infrastructure is key
Intelligent architectural designs frozen within the current paradigm. Data-centric.
MODEL success heavily depends on data.
Biggest leverage in managing and improving data Quality and Quantity.
DATA utilization again is tied to compute.

"We don't have better algorithms. We just have better data." - Norvig (Director Search @Google)

Remember how more data is a regularization method, driving generalization of your models?

LECTURE III

# Data Engineering Fundamentals

# This lecture in one slide

**III Data Engineering Fundamentals**

Data Sources
Data Formats
Data Models
Data Storage Engines and Processing
Modes of Dataflow
Batch Processing Versus Stream Processing

DATA
ENGINEERING
FUNDAMENTALS

Knowing how to handle a growing amount of data

Data Sources
DATA ENGINEERING FUNDAMENTALS
... or knowing how to collect, process, store, retrieve and process an increasing amount of data.

Data Sources
Data Formats, Structures, & Models
DATA ENGINEERING FUNDAMENTALS
...or knowing how to collect, process, store, retrieve and process an increasing amount of data.

Data Sources
Data Formats, Structures, & Models
DATA ENGINEERING FUNDAMENTALS
Data Storage, Retrieval & Processing.
... or knowing how to collect, process, store, retrieve and process an increasing amount of data.

Further reading: Kleppmann 2017

1

# Data Sources

Characteristics, purposes, processing methods

DATA SOURCES

- Characteristics
- Purposes
- Processing Methods

... knowing your data lets you use your data more efficiently.

Common practice to log everything you can.
... and you can store.
... and are allowed to.

Log everything you can, store, are allowed to

Common practice to log everything you can.

... and you can store.

... and are allowed to.

Uses Input Data

Halformatted, any format, requires fast processing.

Common practice to log everything you can.
... and you can store.
... and are allowed to.

Users Input Data
Halformatted, any format, requires fast processing.

System Generated Data
Logs, System outputs, model predictions.

Common practice to log everything you can.

... and you can store.

... and are allowed to.

**User Input Data**

Half-formatted, any format, requires fast processing.

**System Generated Data**

Logs, System outputs, model predictions.

**Internal Databases**

Enterprise applications and assets.

Common practice to log everything you can.

... and you can store.

... and are allowed to.

**User Input Data**

Halfformatted, any format, requires fast processing.

**System Generated Data**

Logs, System outputs, model predictions.

**Internal Databases**

Enterprise applications and assets.

**Third Party Data**

Data collected on the public. As opposed to customers (1st). Apps, Websites, "Public Spaces", "the Internet".

Third party data is usually sold

Third Party Data

"Data Collected on the public. As opposed to customers (1st). Apps, Websites, "Public Spaces, "the Internet".

€

Usually sold after being cleaned and processed.

Data privacy vs. Fingerprinting.

#Apple vs. China.

2

# Data Formats

DATA FORMATS

to pick a format, arouses the question on how the data will be used in future.

To pick a format, ask how the data will be used

DATA FORMATS.

to pick a format, answer the question on how the data will be used in future.

Then match the format based on the access pattern, meaning how a system will read or write your data.

DATA FORMATS.

to pick a format, answer the question on how the data will be used in future.

Then match the format based on the access pattern. meaning how a system will read or write your data.



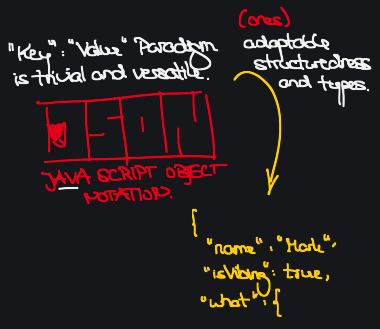

JSON
CSV
PARQUET
Avro
Protobuf
Pickle
!

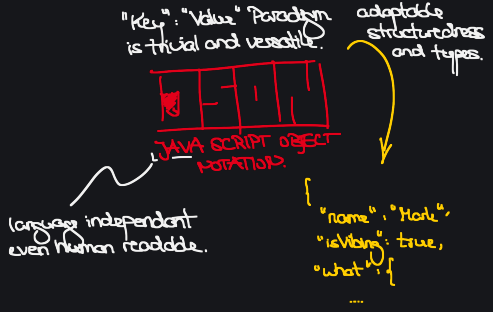
# DATA FORMATS.

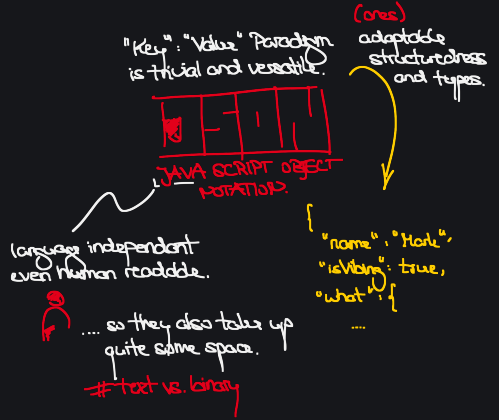
to pick a format, answer the question on how the data will be used in future.

Then match the format based on the access pattern. Meaning how a system will read or write your data.

"Key": "Value" Paradigm
is trivial and versatile.
JSON
JAVA SCRIPT OBJECT
NOTATION


"Key" : "Value" Paradigm
is trivial and versatile.
(ones)
adaptable
structuredness
and types.
JAVA SCRIPT OBJECT
NOTATION.
{
"name" : "Mark",
"isWorking": true,
"what": {
....


"Key" : "Value" Paradigm is trivial and versatile.
(ones) adaptable structuredness and types.
JAVA SCRIPT OBJECT NOTATION.
language independant even human readable.
{
"name" : "Mark",
"isVibing": true,
"what": {
....

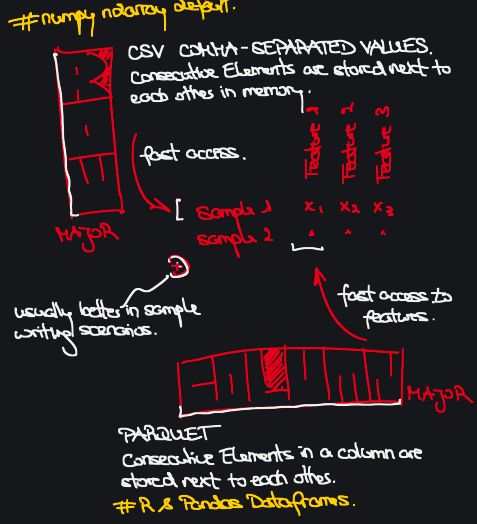

"Key": "Value" Paradigm is trivial and versatile.
(ones) adaptable structuredness and types.
JAVA SCRIPT OBJECT NOTATION.
{
"name": "Mark",
"isWorking": true,
"what": {
....
language independant even human readable.
.... so they also take up quite some space.
#text vs. binary

CSV: consecutive elements side by side in memory

Row major: fast access to samples

Usually better in sample writing scenarios

#numpy ndarray default.
CSV COMMA-SEPARATED VALUES.
Consecutive Elements are stored next to each other in memory.
fast access.
Feature 1
Feature 2
Feature 3
Sample 1
Sample 2
MAJOR
usually better in sample wise processing scenarios.
fast access to features.
MAJOR
PARQUET
Consecutive Elements in a column are stored next to each other.


#numpy ndarray default.
CSV COMMA-SEPARATED VALUES.
Consecutive Elements are stored next to each other in memory.
fast access.
ROW-MAJOR
Feature 1
Feature 2
Feature 3
Sample 1
Sample 2
usually better in sample writing scenarios.
fast access to features.
COLUMN-MAJOR
PARQUET
Consecutive Elements in a column are stored next to each other.
#R & Pandas Dataframes.

Text files: assume each character is one byte

Binary files need knowledge to interpret

Assume,
Each character
is one byte.

Text files

not even one byte but two
characters .. depending on
type.
1000011...

Binary Files
Programs require knowledge
to interpret those.

2x faster to load
6x more compact

3

# Data Models

Data models: how data is represented

Data models affect what your system can solve

RELATIONAL.

each relation is a set of tuples, which are not ordered.



„Think table."

#CSV, Parquet.

NoSQL or nonrelational
or not only relational.
RELATIONAL.
each relation is a
set of tuples, which
are not ordered.
Document Model
Same format: ID.
Think, just rows.
Everything in one place.
„Think table."
# CSV, Parquet.
Graph Model
Relationships modeled
explicitly.

QUICK SIDEQUEST ON QUERY LANGUAGE: SQL.

A declarative language, or „you specify the output you want", not the steps that would take you there (imperative).

DATA MODELS are a pain to maintain, extend or migrate. Unstructured data doesn't adhere to a predefined schema.

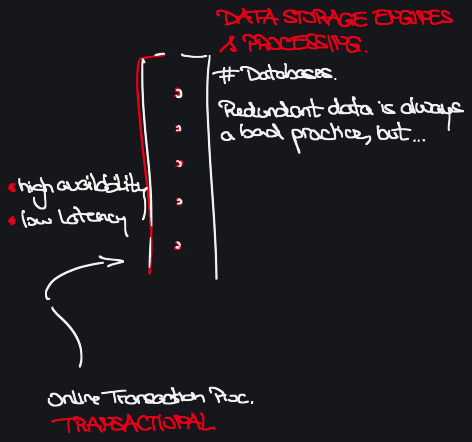
Unstructured data has no predefined schema

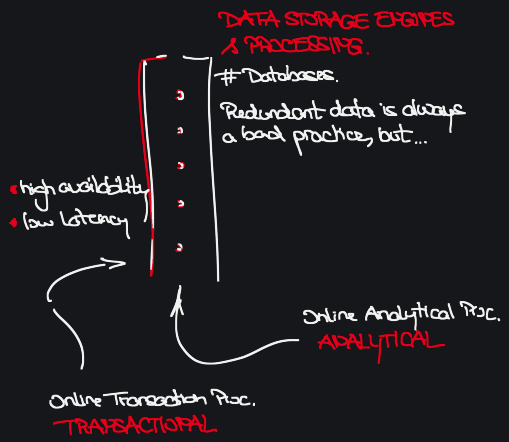
DATA MODELS are a pain to maintain, extend or migrate. unstructured data doesn't adhere to a predefined schema.
But it usually still contains valuable patterns and can be stored effectively #1due.

4

# Data Storage Engines and Processing

Databases: redundant data is a bad practice, but


DATA STORAGE ENGINES & PROCESSING.
#Databases.
Redundant data is always a bad practice, but...
high availability
low latency
Online Transaction Proc.
TRANSACTIONAL


DATA STORAGE ENGINES & PROCESSING.
# Databases.
Redundant data is always a bad practice, but...
high availability
low latency
Online Analytical Proc.
ANALYTICAL
Online Transaction Proc.
TRANSACTIONAL

Online Transaction Proc.

TRANSACTIONAL

Atomicity.

Transaction fails with a single failed step.

Online Transaction Proc.

TRANSACTIONAL

- **Atomicity.**
  Transaction fails with a single failed step.
- **Consistency.**
  All transactions follow the same rules

Online Transaction Proc.

TRANSACTIONAL

**Atomicity.**
Transaction fails with a single failed step.

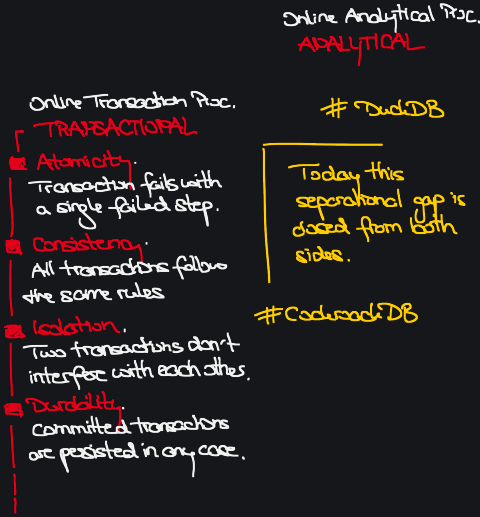
**Consistency.**
All transactions follow the same rules

**Isolation.**
Two transactions don't interfere with each other.

Online Transaction Proc.

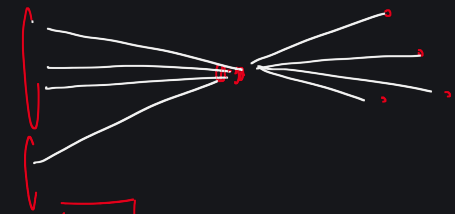
TRANSACTIONAL

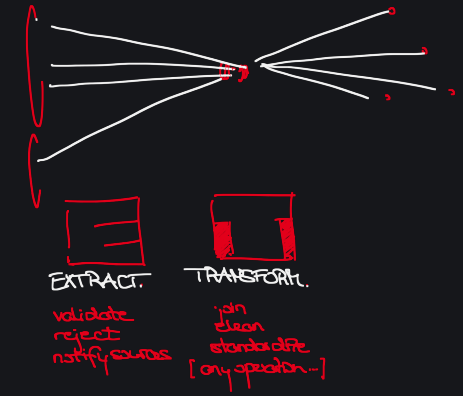
1. Atomicity.
   Transaction fails with a single failed step.
2. Consistency.
   All transactions follow the same rules
3. Isolation.
   Two transactions don't interfere with each other.
4. Durability.
   Committed transactions are persisted in any case.


Online Transaction Proc.
TRANSACTIONAL
Atomicity.
Transaction fails with a single failed step.
Consistency.
All transactions follow the same rules
Isolation.
Two transactions don't interfere with each other.
Durability.
Committed transactions are persisted in any case.
Online Analytical Proc.
ANALYTICAL
#DuckDB
Today this separational gap is closed from both sides.
#CockroachDB

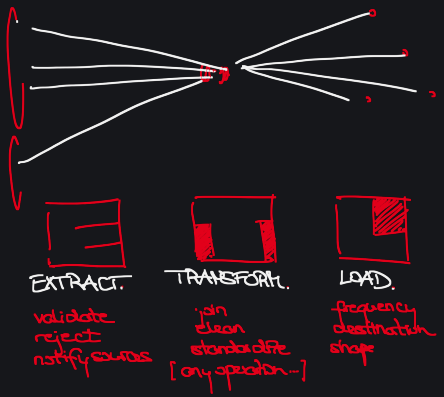
Paradigm to decouple storage from compute,
to reach general purpose data.

Paradigm to decouple storage from compute, to reach general purpose data.

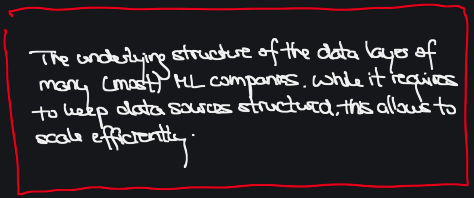



Paradigm to decouple storage from compute,
to reach general purpose data.
EXTRACT.
validate
reject
notify issues
TRANSFORM.
join
clean
standardize
[any operation...]

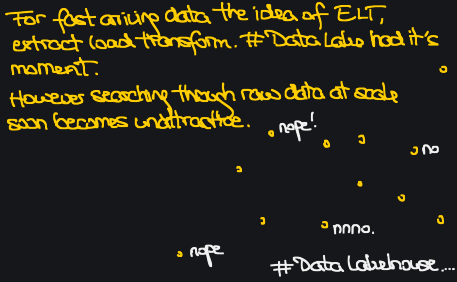
Paradigm to decouple storage from compute, to reach general purpose data.



The structure of most ML data layers



For fast arriving data the idea of ELT, extract load transform. #Data Lake had it's moment.
However searching through raw data at scale soon becomes unattractive.


For fast arriving data, the idea of ELT, extract load transform. #Data Lake had it's moment.
However searching through raw data at scale soon becomes unattractive.
nope!
no
nnnno.
nope
#Data Lakehouse....

5

# Modes of Dataflow

Not all data is persisted, not all memory shared

Write to a database: persisted and shared

Read from the database, slow

SERVICE-ORIENTED DESIGN.
Not all data is persisted.
Not all processes share memory.
Use a shared network.
Process B
Process A

Request and expose

More processes, more requests

Microservice Architecture
as a design principle.
Develop, test, maintain
independently. #API-first. #REST

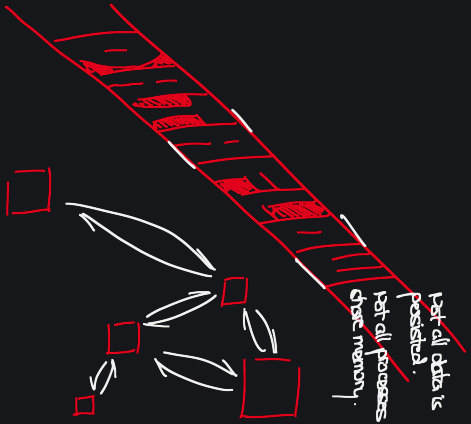
Microservice Architecture
as a design principle.
Develop, test, maintain
independently. #API first. #REST

FURTHER READING:
Building Microservices by Sam Newman.

Microservice Architecture
as a design principle.
Develop, test, maintain
independently. #API-first. #REST

FURTHER READING:
Building Microservices by Sam Newman.

... and then design your org chart around your architecture.




Many processes, many data paths

Passing data gets complicated and stale fast

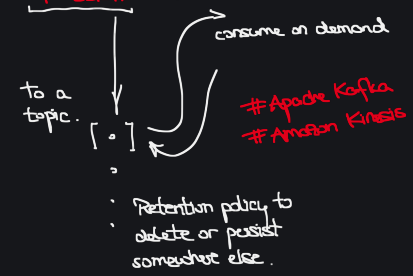
EVENT-DRIVEN.
Not all data is persisted.
Not all processes share memory.
DATAFLOW

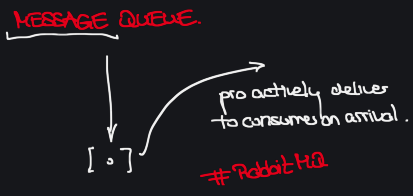
Processes publish events to a bus

The broker, usually in memory

Publish subscribe: to a topic, consume on demand

Retention policy. Apache Kafka, Amazon Kinesis


PUBLISH SUBSCRIBE.
consume on demand
to a topic.
#Apache Kafka
#Amazon Kinesis
Retention policy to delete or persist somewhere else.


MESSAGE QUEUE.
pro actively deliver to consumers on arrival.
#RabbitMQ

6

# Batch Processing Versus Stream Processing

This is all for streaming data, as opposed to historical, batch processed data.

This is the last slide.
Look how far you've come.

# References


1 Chip Huyen. *Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications*. Sebastopol, CA: O'Reilly Media, 2022.

2 McKinsey & Company. *The State of AI: Global Survey 2026*. Tech. rep. McKinsey & Company, QuantumBlack, 2026.

3 AI Index Steering Committee. *The AI Index 2026 Annual Report*. Tech. rep. Institute for Human-Centered AI, Stanford University, 2026.

4 Kate Niederhoffer et al. "AI-Generated "Workslop" Is Destroying Productivity". In: *Harvard Business Review* (Sept. 2025).

5 Dario Amodei. *CEO Speaker Series With Dario Amodei of Anthropic*. Council on Foreign Relations, New York, 10 March 2025, transcript. 2025.

6 Anthropic. *Open Roles, Software Engineer*. Anthropic job board on Greenhouse, read 18 October 2026. 2026.

7 Akamai Technologies. *State of Online Retail Performance, Spring 2017*. Tech. rep. Akamai Technologies, 2017.

8 Lukas Vermeer. *Moving Fast: Latency and Conversion at Booking.com*. Booking.com engineering. 2019.

9 Google. *The Need for Mobile Speed: How Mobile Latency Impacts Publisher Revenue*. Tech. rep. Google and DoubleClick, 2016.

10 Betsy Beyer et al. *Site Reliability Engineering: How Google Runs Production Systems*. Sebastopol, CA: O'Reilly Media, 2016.

# References


11 David Picard. *torch.manual_seed(3407) is All You Need: On the Influence of Random Seeds in Deep Learning Architectures for Computer Vision*. arXiv:2109.08203. 2021.

12 Kaggle. *Titanic: Machine Learning from Disaster*. Kaggle getting started competition. 2012.

13 Andrej Karpathy. *Building the Software 2.0 Stack*. Train AI 2018, San Francisco. 2018.

14 Alex Wang et al. "GLUE: A Multi-Task Benchmark and Analysis Platform for Natural Language Understanding". In: *Proceedings of the 2018 EMNLP Workshop BlackboxNLP*. 2018, pp. 353–355.

15 Inioluwa Deborah Raji et al. "AI and the Everything in the Whole Wide World Benchmark". In: *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*. 2021.

16 Timo Freiesleben and Sebastian Zezulka. *The Benchmarking Epistemology: Construct Validity for Evaluating Machine Learning Models*. arXiv:2510.23191. 2025.

17 Lianmin Zheng et al. "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena". In: *Advances in Neural Information Processing Systems 36*. 2023.

# References


18 Arjun Panickssery, Samuel R. Bowman, and Shi Feng. “LLM Evaluators Recognize and Favor Their Own Generations”. In: *Advances in Neural Information Processing Systems 37*. 2024.

19 Dawei Li et al. *Preference Leakage: A Contamination Problem in LLM-as-a-judge*. arXiv:2502.01534. 2025.

20 Jeffrey Dastin. *Amazon Scraps Secret AI Recruiting Tool That Showed Bias Against Women*. Reuters, 10 October 2018. 2018.

21 Tom Simonite. *When It Comes to Gorillas, Google Photos Remains Blind*. Wired, 11 January 2018. 2018.

22 Moritz Hardt. *The Invisible Hand of Prediction*. Simons Institute for the Theory of Computing, Berkeley. 2022.

23 Juan C. Perdomo et al. “Performative Prediction”. In: *Proceedings of the 37th International Conference on Machine Learning*. Vol. 119. PMLR. 2020, pp. 7599–7609.

24 TypeSafe AI. *Jev: A Model That Returns Decisions, Not Text*. Model launch, early access, 15 September 2026. 2026.

25 National Transportation Safety Board. *Collision Between Vehicle Controlled by Developmental Automated Driving System and Pedestrian, Tempe, Arizona, March 18, 2018*. Highway Accident Report NTSB/HAR-19/03. Washington, DC: National Transportation Safety Board, 2019.

# References


26 Titus Winters, Tom Manshreck, and Hyrum Wright. *Software Engineering at Google: Lessons Learned from Programming Over Time*. Sebastopol, CA: O'Reilly Media, 2020.

27 Frederick P. Brooks. *The Mythical Man-Month: Essays on Software Engineering*. Reading, MA: Addison-Wesley, 1975.

28 Martin Kleppmann. *Designing Data-Intensive Applications*. Sebastopol, CA: O'Reilly Media, 2017.

29 Sam Newman. *Building Microservices: Designing Fine-Grained Systems*. 2nd ed. Sebastopol, CA: O'Reilly Media, 2021.